Punjab Canal Regulation Information System

(PCRIS, Version 1.0)

User Manual

Designed & Developed by

National Informatics Centre, Punjab

Introduction – Punjab Canal Regulation

Punjab, the name derived from the Persian words, Punj (five) and ab (water) was a land of five rivers which formed part of Indus Basin till 1947. After partition of the country, Indus water Treaty-1960 restricted India's rights of usage to only three Eastern rivers (Sutlej, Ravi and Beas), the three Western rivers (Indus, Chenab and Jhelum) were earmarked for exclusive usage of Pakistan. Punjab does have a very well developed and interlinked river system and widespread 14500 km long Canal Systems. Water is most essential natural resource next to the air, a basic human need and the most important input for all human development activities and, perhaps, very precious and scarce natural resource. Efforts need to be made to develop, plan, conserve, utilize and manage this important resource, both surface and ground water, in a judicious, equitable, sustainable and sound economic manner.

The Information System -

Home Page: Punjab Canal Regulation Information System (PCRIS) is a web based application for managing entire canal network in Punjab. Application is accessible using URL https://pcris.punjab.gov.in. The application welcomes its users with an informative screen containing tabs showing various water entities like Dam/ Headworks, Canals etc.

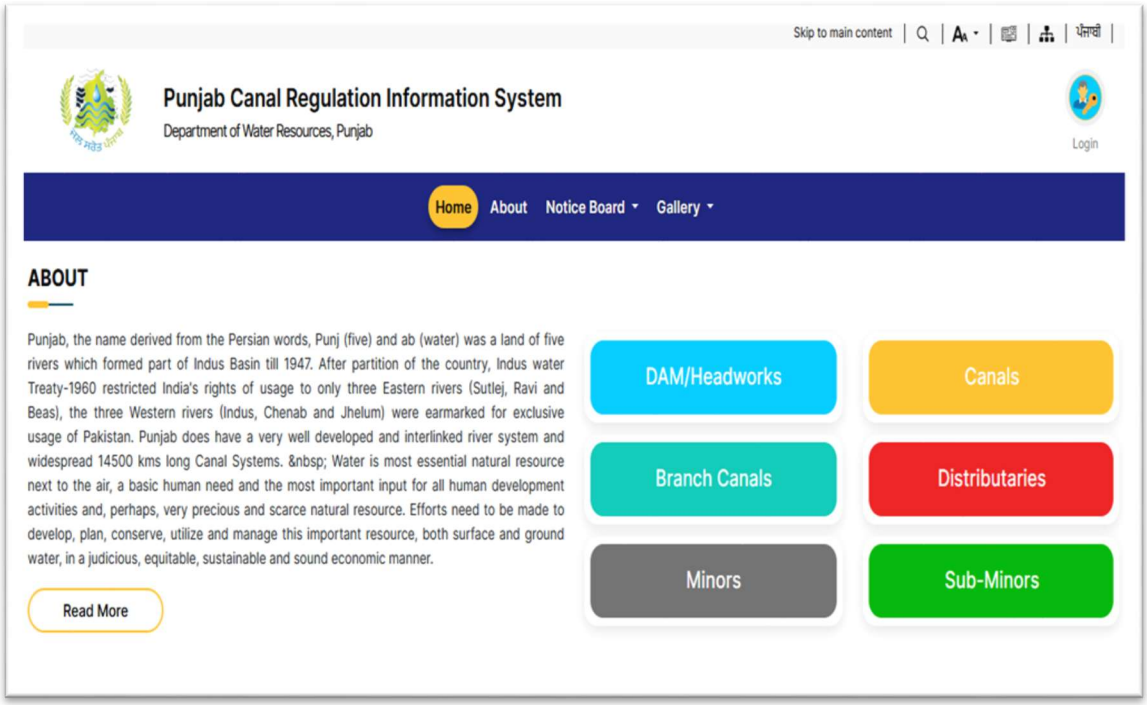


Figure 1: PCRIS Welcome Screen

Scrolling down the homepage will present a bharatmaps view with geospatial information

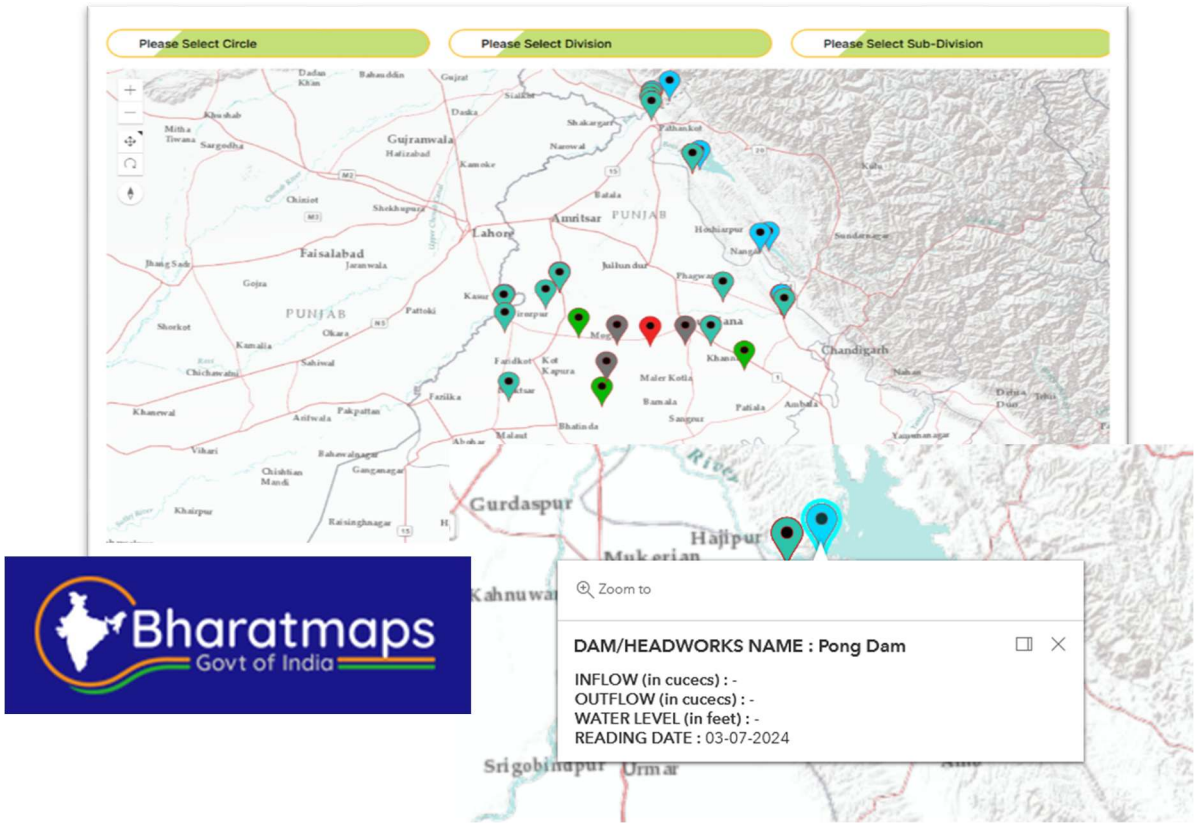


Figure 2: Bharatmaps, Balloon Icons and Water Entity Details

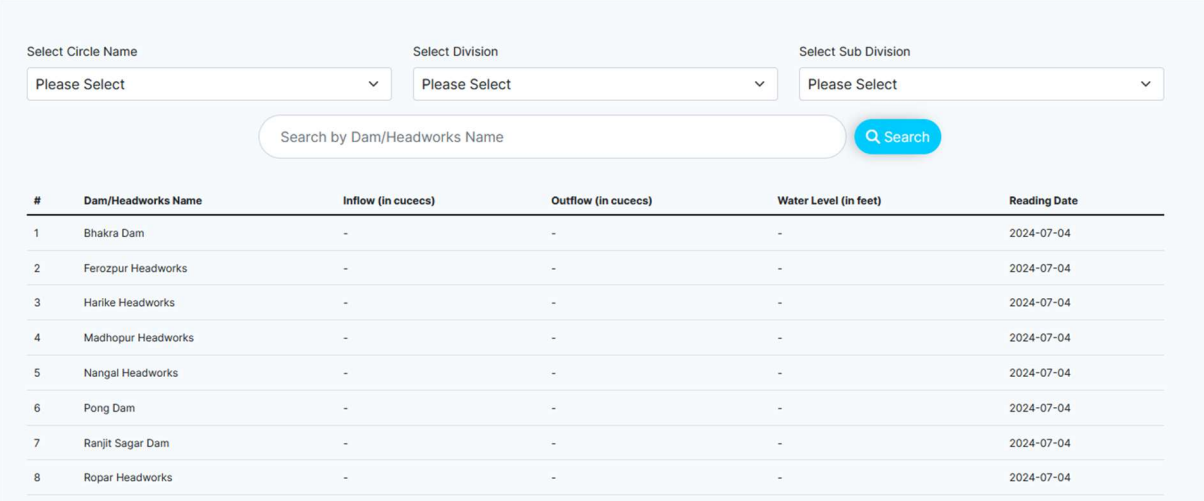
of all water entities, using balloon icons with same color as tab button color. Apart from this, many other useful information is also there on welcome screen. Clicking the balloons will display a brief but useful information about that water entity.

By default bharatmaps will display all the water entities, those are entered in application with latitude and longitude information. Filters are also provided for bharatmaps through which display of bharatmaps can be controlled. Water entities on bharatmaps can be filtered by selecting Circle/ Division/ Sub Division. *Scrolling mouse over bharatmaps will zoom-in or zoom-out the display.*



Figure 3: Filters - Circle, Division and Sub Division

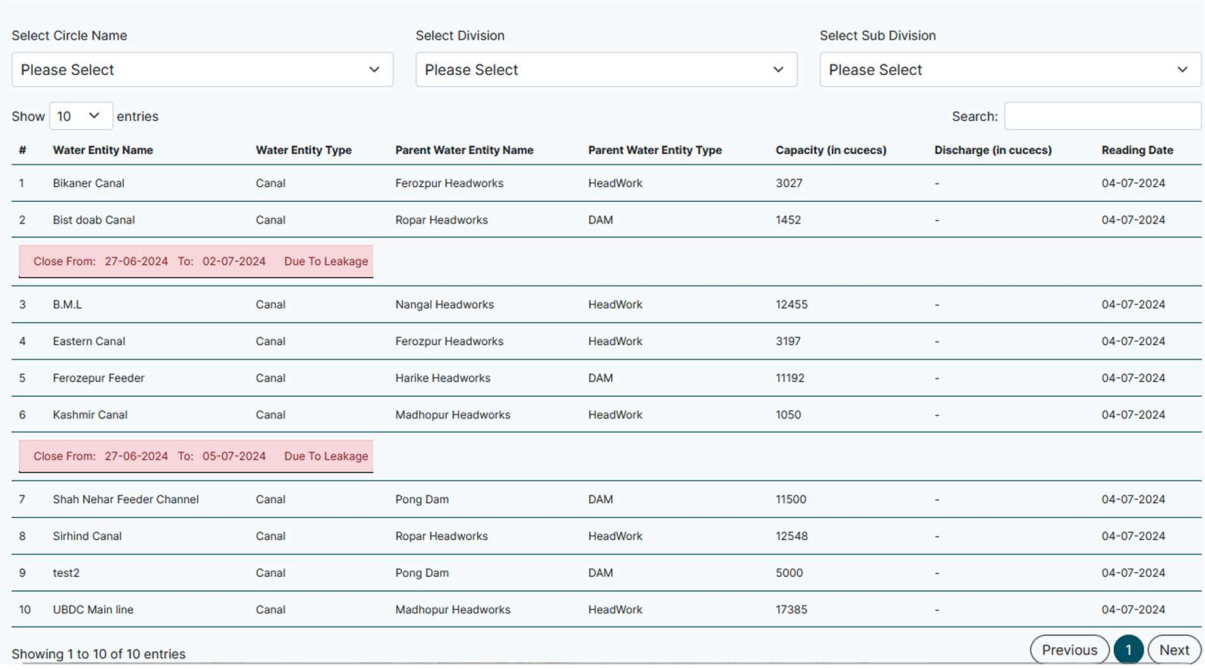
Clicking any tab button shown in *Figure-1*, will display a detailed information of related water entity with filters. Information is displayed in a tabular view making it more understandable. For example, clicking Dam/ Headwork button will display below information:



#	Dam/Headworks Name	Inflow (in cucecs)	Outflow (in cucecs)	Water Level (in feet)	Reading Date
1	Bhakra Dam	-	-	-	2024-07-04
2	Ferozpur Headworks	-	-	-	2024-07-04
3	Harike Headworks	-	-	-	2024-07-04
4	Madhopur Headworks	-	-	-	2024-07-04
5	Nangal Headworks	-	-	-	2024-07-04
6	Pong Dam	-	-	-	2024-07-04
7	Ranjit Sagar Dam	-	-	-	2024-07-04
8	Ropar Headworks	-	-	-	2024-07-04

Figure 4: Information regarding Dam/ Headwork

From this table one can have a details of inflow, outflow and water level and information can be filtered to a specific Circle, Division and sub Division. Search box is also provided, user can type in name of water entities to be specific. Similarly, following is the information related to Canal Button.



#	Water Entity Name	Water Entity Type	Parent Water Entity Name	Parent Water Entity Type	Capacity (in cucecs)	Discharge (in cucecs)	Reading Date
1	Bikaner Canal	Canal	Ferozpur Headworks	HeadWork	3027	-	04-07-2024
2	Bist doab Canal	Canal	Ropar Headworks	DAM	1452	-	04-07-2024
	Close From: 27-06-2024 To: 02-07-2024 Due To Leakage						
3	B.M.L	Canal	Nangal Headworks	HeadWork	12455	-	04-07-2024
4	Eastern Canal	Canal	Ferozpur Headworks	HeadWork	3197	-	04-07-2024
5	Ferozepur Feeder	Canal	Harike Headworks	DAM	11192	-	04-07-2024
6	Kashmir Canal	Canal	Madhopur Headworks	HeadWork	1050	-	04-07-2024
	Close From: 27-06-2024 To: 05-07-2024 Due To Leakage						
7	Shah Nehar Feeder Channel	Canal	Pong Dam	DAM	11500	-	04-07-2024
8	Sirhind Canal	Canal	Ropar Headworks	HeadWork	12548	-	04-07-2024
9	test2	Canal	Pong Dam	DAM	5000	-	04-07-2024
10	UBDC Main line	Canal	Madhopur Headworks	HeadWork	17385	-	04-07-2024

Figure 5: Information using Canals button

Here also filters as well as search box are provided. If water entities is being closed for any reason the same is displayed as Red warning message. Previous and Next buttons can be used to traverse among pages if information is spread among more than one pages.

In the top line on home screen a few icons are available. Their functionalities are described briefly as following:

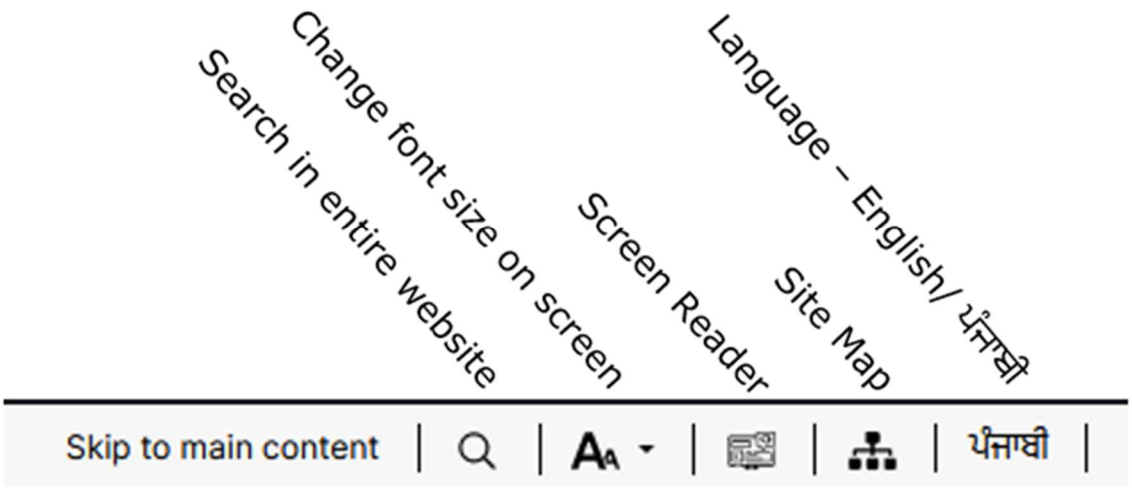


Figure 6: Top row icons

- **Skip to main content** – Clicking this will quickly jumps you to the starting of the main contents on that page skipping the rest of the headers and other information.
- **Search in entire website** – Type in text wants to search. As a result it will display all the pages in which it find the desired text.
- **Change font size on screen** – Use it to increase or decrease the on screen text size.
- **Screen Reader** - PCRIS platform will enable people with visual impairments access the website using assistive technologies, such as screen readers. The information of the website is accessible with different screen readers, such as NVDA.
- **Site Map –** It will provide information about the pages, videos, and other files on your site, and the relationships between them.
- **Language, English/ Punjabi (ਪੰਜਾਬੀ)** – Click it to change language of information.

  (The contents will be available if department has provided in the selected language.)

Beyond Welcome Screen-

This welcome screen serves its purpose equally for official and general public. In order to access other functionalities related to official work, officials have to use login option present on the right top welcome screen.

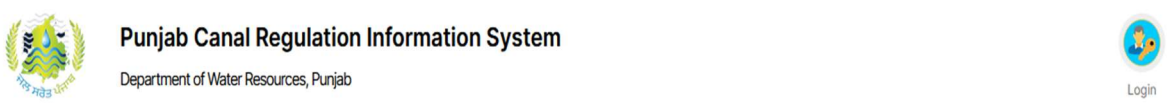


Figure 7: Login option

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